

Nilardri Pramanick

Kolkata, West Bengal, India

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Summary

Third-year B.Tech Information Technology student specializing in AI, Machine Learning, and Agentic Systems. Experienced building multi-agent LLM pipelines with LangGraph/LangChain, ensemble deep learning models, and full-stack AI applications. IEEE research intern in NLP/sentiment analysis; hackathon builder (NASA Space Apps, Google Cloud GenAI Exchange) with strong DSA and production-ready system design skills. Seeking AI Engineer / ML Engineer roles.

Technical Skills

- **Languages:** Python, C++, C, JavaScript, SQL
- **AI / ML / Generative AI:** LangGraph, LangChain, LangSmith, Agentic AI Workflows, Multi-Agent Systems, Agent Orchestration, LLM Evaluation (Ragas, DeepEval), PyTorch, TensorFlow, Hugging Face, OpenAI API, Cohere, Groq, Ensemble Learning, NLP, Deep Learning (CNN, Transformers), Explainable AI (SHAP), Prompt Engineering, RAG
- **Backend & Full-Stack:** Django, FastAPI, Flask, DRF, React.js, HTML, CSS, REST API Development
- **Databases & Cloud:** MongoDB, PostgreSQL, SQL, Firebase, Cloudinary
- **Data & Tools:** NumPy, Pandas, Matplotlib, Seaborn, Streamlit, Git, GitHub, JWT Authentication
- **Core CS:** Data Structures & Algorithms (Graphs, Dynamic Programming), Computer Architecture, Digital Electronics (VHDL)

Experience

IEEE Research Intern

Mentor: Dr. Sujoy Chatterjee

- Researched **ensemble sentiment analysis** for real-time, event-driven public opinion mining from social media data.
- Built and benchmarked multiple NLP models – **VADER, RoBERTa, BERT-Multilingual, DistilBERT** – and combined outputs via an **ensemble learning** framework to improve accuracy and interpretability.
- Authored a technical report, “Understanding Public Opinion from Social Media: A Deep Dive into Ensemble Sentiment Analysis,” evaluating model accuracy, confidence, and robustness to noisy, fast-changing social data.

Projects

NextHire – Voice-Based AI Interview Practice Platform *LangGraph, FastAPI, Streamlit, PostgreSQL, Cohere, Faster-Whisper, Kokoro TTS*

github.com/Nilardri2006-2/NextHire

- Designed a **stateful LangGraph agent pipeline** (resume/policy ingestion, adaptive screening, evaluation, report generation) with structured output extraction via Pydantic schemas for reliable question and policy parsing from uploaded PDFs.
- Built a full **voice conversational loop** using Faster-Whisper (speech-to-text) and Kokoro (text-to-speech), enabling spoken, non-repeating interview questions grounded dynamically in the candidate’s resume and interview history.
- Implemented an **automated LLM-based evaluation engine** scoring candidates across 10 weighted criteria (communication, technical understanding, role fit, etc.) with evidence-grounded justifications, and auto-generated PDF interview reports via ReportLab.
- Exposed the pipeline as a **FastAPI REST backend** with a Streamlit front-end, and added **Postgres-backed state checkpointing** for crash-resilient, resumable interview sessions.

AI-itinerary – LangGraph Multi-Agent Travel Planner

Python, LangGraph, LangChain, Tavily, React

github.com/Nilardri2006-2/AI-itinerary-v3

- Built a **multi-agent AI trip planner**, deployed live on Vercel, that converts a single natural-language prompt into a complete itinerary via a 6-agent LangGraph pipeline with MCP (Flight, Itinerary, Transport, Hotel, Image agents, Weather agents) using Tavily search and the Wikipedia API for live hotel discovery and imagery.

AI-Powered Automotive Warranty Claim Adjudication System *Python, LangGraph, LangChain, Cohere, Streamlit*

github.com/Nilardri2006-2/fraud-detector

- Developed a **4-agent LangGraph decision-support system** (Policy Check, Fraud Scoring, Evidence Collector, Decision Agent) that validates policy compliance (coverage, mileage, warranty period), scores fraud likelihood, and outputs an explainable Approve / Reject / Escalate recommendation with human-in-the-loop (HITL) escalation for ambiguous cases.

Hackathons & Competitions

- **NASA Space Apps Challenge 2025 – Team Ares:** Built **ATLAS**, an AI exoplanet-detection pipeline on Kepler/K2/TESS data using CNNs, Transformers, ensemble models, and SHAP explainability.
- **Google Cloud GenAI Exchange Hackathon 2025:** Built **Indus AI**, a generative-AI-powered solution developed within a limited hackathon timeframe.
- **Hack0Solution Hackathon:** Built **Swasthya**, a GenAI medical insurance summarizer that parses insurance documents into plain-language summaries of coverage and claims. github.com/bytebrigade2006/Swasthya

Leadership & Involvement

- **Founding Member, Byte Brigade** – core founding team member of a student tech community, collaborating on peer projects.
- **Member, Cyber Nix** – active member of NSEC’s cybersecurity/tech wing.
- **PR Member, IEEE NSEC Student Chapter** – handled public relations and outreach for the student chapter.

Education

Netaji Subhash Engineering College (NSEC) <i>B.Tech in Information Technology</i> <i>CGPA till fourth semester -8.0</i>	Kolkata, India <i>Expected 2028</i>
Garden Reach Mudiali High School <i>Higher Secondary – 79%</i>	<i>2024</i>
Languages Spoken: English (Professional) , Bengali (Native), Hindi	